Routines Page — Information Document

Page Overview

Purpose

The Routines page is the management surface for scheduled, automatic sweep jobs. A routine bundles a saved sweep configuration (repositories, keywords, optional date range, max-results cap) with a 5-field cron expression, plus per-routine flags for email, AI summarization, and desktop completion notifications. When the schedule fires, the app runs an `automated_sweep` execution against the configured repositories, stores the report, and (optionally) emails it and/or raises a desktop notification.

Routines may be pinned to either of two execution locations: **Local** (fired by the APScheduler background thread inside the local `resmon` daemon) or **Cloud** (fired by the `resmon-cloud` scheduler, via the JWT-gated `/api/v2/routines` service). Both kinds are rendered side-by-side in a single unified table.

Primary User Flows

1. **Create a routine.** Open the Create modal, optionally load a saved routine configuration via `ConfigLoader`, pick repositories, keywords, an optional date range, a max-results cap, flags (AI / Email / Results-in-Email / Notify-on-Completion), a cron expression, and an execution location. Submit to persist.
2. **Edit an existing routine.** Click **Edit** on a local row. The modal pre-populates from the existing row; saving issues `PUT /api/routines/{id}`.
3. **Activate / deactivate.** Toggle scheduling without deleting via the Activate / Deactivate action button. Deactivation removes the APScheduler job but preserves the DB row.
4. **Per-row quick toggles.** The Email, AI, and Notify columns expose single-click toggles that patch the matching flag with `PUT /api/routines/{id}`.
5. **Migrate between locations. Move to Cloud** and **Move to Local** trigger a confirmation modal, then perform a two-step destination-first create / source-delete migration.
6. **Cancel a live run.** When a routine is currently firing, a **Cancel Run** button appears on its row and calls through `ExecutionContext.cancelExecution`.

7. **Delete a routine.** Delete removes the DB row (local) or the cloud record; historical execution rows produced by that routine are retained. **Update 4 (`5_5_26`):** `delete_routine` additionally removes the matching `apscheduler_jobs` row in the same transaction so a deleted routine never leaves a ghost job behind, and the daemon's scheduler bootstrap performs a one-shot reconciliation that drops every `apscheduler_jobs` row whose id has no matching active `routines.id` . Routine jobs are also registered with `misfire_grace_time=3600` so a fire whose nominal time briefly passed (daemon restart, scheduler reattach, or a renderer-spawned scheduler dying without `shutdown()`) still runs instead of being silently dropped.
8. **Materialized via Configurations import.** Importing a routine config on the Configurations page automatically materializes a matching local routine row on this page (`is_active=False` by default, so the imported routine is deactivated until the user explicitly activates it). The page subscribes to the configurations bus (`useConfigurationsVersion()`) so newly-imported routines surface here without a manual reload. This keeps the Routines list and the routine-configs list in lockstep — every routine config has a real routine, and vice-versa.

Inputs and Outputs

Inputs:

- Routine name, cron expression, repositories, optional date range, keywords, max-results cap.
- Flags: `is_active` , `email_enabled` , `email_ai_summary_enabled` , `ai_enabled` , `notify_on_complete` , `execution_location` .
- Optional `ai_settings` and `storage_settings` payloads (stored as JSON; not edited directly on this page).

Outputs:

- Persisted `routines` rows (local) or `routines` documents on the cloud scheduler.
- APScheduler jobs registered under the routine's id as `job_id` .
- Automated `executions` rows produced at each fire, stamped with `routine_id` and `last_executed_at` .
- Downstream report artifacts (Markdown / PDF / figures) under `resmon_reports/` .

Known Constraints or Permissions

- Cloud rows render only when the user is signed in (`useAuth().isSignedIn`); cloud list failures are swallowed and the table simply shows the local half.
- The **Execution location** → **Cloud** radio is disabled unless cloud sync is enabled (signed in).
- Cron is 5-field (`m h dom mon dow`); the default value in the create form is `0 8 * * *` .

- Only local rows expose the Edit, Email/AI/Notify toggles, Activate/Deactivate, Last Execution, and Last Status columns — cloud rows show em-dashes in those columns because cloud state is not mirrored into the local DB for rendering.
- Migration atomicity is orchestrated renderer-side: destination is created first; source is deleted only if that succeeded.

Frontend

Route and Main Component

- Route: `/routines`.
- Main component: `resmon_scripts/frontend/src/pages/RoutinesPage.tsx`.

Child Components and Hooks

Child components:

- `RoutineEditModal` (from `components/Routines/RoutineEditModal.tsx`) — the create/edit modal has been extracted into a reusable component shared with the Calendar page (whose event popover exposes an `Edit Routine` button). On save it broadcasts BOTH `notifyRoutinesChanged()` (via `lib/routinesBus.ts`) and `notifyConfigurationsChanged()` (via `lib/configurationsBus.ts`) so every routine-aware surface refetches without a manual reload. The host pages own only `editTarget: Routine | null` and an `editOpen` flag; all form state, validation, and the inner editor children below live inside the modal.
- `RepositorySelector` (multi-select repository picker).
- `KeywordCombinationBanner` — mounted in the create/edit modal under the selector; lists the upstream keyword-combination semantics for each selected repository, with a tooltip pointing to the consolidated glossary on the Repositories & API Keys page.
- `DateRangePicker`, `KeywordInput`, `ScheduleConfigurator` (cron builder).
- `ConfigLoader` with `configType="routine"` — populates everything except the date range, which is re-picked per run.
- `RepoKeyStatus` with `variant="routine"` — one row per selected repository, showing credential presence.
- `PageHelp` and `InfoTooltip` — help UI only.

Hooks:

- `useAuth()` — source of `isSignedIn`, gating cloud rows and the Cloud execution-location radio.
- `useExecution()` — supplies `activeExecutions`, `cancelExecution`, and `completionCounter`. `completionCounter` is a dependency of both fetchers so the table refreshes when an execution finishes.
- `useRepoCatalog()` — `bySlug` / `presence` / `refreshPresence` back the per-row `RepoKeyStatus` entries.
- `useRoutinesVersion()` (from `lib/routinesBus.ts`) — dependency of the local and cloud fetch effects so saves originating from the Calendar page's `Edit Routine` modal (or any other future surface) refresh the table here.

API clients:

- `apiClient` for the local `/api/routines` surface.
- `cloudClient` for the cloud `/api/v2/routines` surface.

UI State Model

Top-level state:

- `routines: Routine[]` — local rows from `GET /api/routines`.
- `cloudRoutines: CloudRoutineRow[]` — cloud rows from `GET /api/v2/routines`.
- `loading: boolean`, `error: string`.
- `pendingMigration: PendingMigration` — `{ kind: 'to-cloud' | 'to-local', routine: UnifiedRoutine } | null`; drives the confirm modal.
- `migrating: boolean` — disables the confirm modal's buttons while the two-step migration runs.

Create/edit form state:

- `formOpen`, `editId` (`null` = creating, number = editing a local id).
- `formName`, `formCron` (default `"0 8 * * *"`), `formRepos`, `formDateFrom`, `formDateTo`, `formKeywords`, `formMaxResults` (default `100`).
- `formAi`, `formEmail`, `formEmailAi`, `formNotify`, `formLocation` (`'local' | 'cloud'`).

Types:

- `Routine` — local row shape.
- `CloudRoutineRow` — `{ routine_id, name, cron, enabled, parameters }` as returned by `/api/v2/routines`.

- `UnifiedRoutine` extends `Routine` with `execution_location` and optional `cloud_id`. It is the shape handed to the migration modal so both row types share a single confirm path.

Helpers:

- `lastStatusBadgeClass(status)` maps status → badge class: `completed → badge-success`, `failed → badge-error`, `cancelled → badge-cancelled`, everything else (`running`, `cancelling`, `scheduled`, `unknown`) → `badge-info`. This mirrors the Dashboard and Results & Logs palette.

Key Interactions and Events

- **Fetch.** `fetchRoutines` → `GET /api/routines`; `fetchCloudRoutines` → `GET /api/v2/routines` (no-op when `cloudSyncEnabled` is false). Both run on mount and every time `completionCounter` changes.
- **Create.** `handleSubmit` when `editId === null`:
  - If `formLocation === 'cloud'` and signed in: `POST /api/v2/routines` with `{ name, cron, parameters, enabled: true }`.
  - Otherwise: `POST /api/routines` with the full local body (flags + `execution_location: 'local'`).
- **Edit.** `handleSubmit` when `editId !== null`: `PUT /api/routines/{editId}` with name, cron, parameters, `is_active: true`, all flag fields, and `execution_location`.
- **Delete.** `handleDelete(id)` → `DELETE /api/routines/{id}` (local); `handleDeleteCloud(cloudId)` → `DELETE /api/v2/routines/{cloudId}` (cloud).
- **Toggle active.** `handleToggleActive` calls `POST /api/routines/{id}/activate` or `POST /api/routines/{id}/deactivate` based on current `is_active`.
- **Quick toggles.** `handleToggleEmail`, `handleToggleAi`, `handleToggleNotify` each call `PUT /api/routines/{id}` with only the inverted flag in the body.
- **Move to Cloud.** `performMoveToCloud`: `POST /api/v2/routines` (cloud create), then `POST /api/routines/{id}/released-to-cloud` (local delete).
- **Move to Local.** `performMoveToLocal`: `POST /api/routines/adopt-from-cloud` (local create), then `DELETE /api/v2/routines/{cloud_id}` (cloud delete).
- **Cancel running fire.** For each local row, the render scans `activeExecutions` for `executionType === 'automated_sweep'` with matching `routine_id` and status `running` or `cancelling`; if present, a **Cancel Run** button calls `cancelExecution(running.executionId)` and shows a `Stopping…` state while `cancelling`.

Error and Empty States

- **Loading.** `if (loading)` returns a muted "Loading routines…" placeholder.
- **Empty.** When both `routines` and `cloudRoutines` are empty, the table body shows a single "No routines configured." row spanning all 10 columns.
- **Errors.** A non-empty `error` string renders a `form-error` banner above the table. Cloud fetch errors are swallowed intentionally (sign-in dialogs / toasts elsewhere handle auth failures).
- **Missing cloud_id on move-to-local.** `performMoveToLocal` sets `"Missing cloud_id for migration"` and returns without calling any endpoint.
- **Partial migration failures.** If the destination create succeeds but the source delete fails, the error is surfaced and both rows remain; retrying the delete on the second attempt recovers without losing the destination copy (per the renderer-side atomicity comment in the component).

Backend

API Endpoints

Local (`resmon_scripts/resmon.py`):

- `GET /api/routines` — list, enriched with `last_execution` (from `last_executed_at`) and `last_status` (most recent `executions` row by `routine_id`).
- `POST /api/routines` — create (body: `RoutineCreate`).
- `PUT /api/routines/{routine_id}` — partial update (body: `RoutineUpdate` , every field optional).
- `DELETE /api/routines/{routine_id}` — delete.
- `POST /api/routines/{routine_id}/activate` — set `is_active=1` .
- `POST /api/routines/{routine_id}/deactivate` — set `is_active=0` .
- `POST /api/routines/{routine_id}/released-to-cloud` — delete the local row after a successful cloud mirror.
- `POST /api/routines/adopt-from-cloud` — insert a local row populated from a cloud body (preserves name, cron, parameters verbatim).

Cloud (`resmon-cloud` , consumed via `cloudClient`):

- `GET /api/v2/routines` — list cloud routines (`CloudRoutineRow[]`).
- `POST /api/v2/routines` — create a cloud routine (`{ name, cron, parameters, enabled }`).
- `DELETE /api/v2/routines/{routine_id}` — delete a cloud routine.

Request and Response Patterns

- Local request bodies follow the Pydantic models `RoutineCreate` and `RoutineUpdate`. `RoutineCreate` requires `name`, `schedule_cron`, `parameters`; defaults `is_active=True`, all flags `False`, and `execution_location="local"`. `RoutineUpdate` makes every field optional and only applies the keys the renderer actually sends.
- `execution_location` is validated to be `'local'` or `'cloud'`; anything else returns `400`.
- Create returns `{id, name}` with status `201`.
- Update returns `{id, ...updates}`.
- `released-to-cloud` returns `{released: True, id}`; `adopt-from-cloud` returns `{id, name, execution_location: "local"}` with status `201`.
- Cloud requests use the dedicated `cloudClient` wrapper so JWT headers (IMPL-30) are applied; cloud bodies use the short `cron` / `enabled` / `routine_id` field set.

Persistence Touchpoints

- SQLite `routines` table is the authoritative store for local rows (written via `insert_routine`, `update_routine`, `delete_routine`, `get_routines`, `get_routine_by_id`).
- `parameters`, `ai_settings`, and `storage_settings` are serialized as JSON strings. Integer-encoded booleans are used for `is_active`, `email_enabled`, `email_ai_summary_enabled`, `ai_enabled`, and `notify_on_complete`.
- `executions` rows produced by a routine fire carry the FK `routine_id`; the list endpoint joins the most recent such row to expose `last_status`.
- APScheduler job state lives in a separate SQLAlchemy job store (SQLite) managed by `ResmonScheduler`; job id is the string form of the routine id.
- `_sync_routine_config` / `_delete_routine_config` keep the per-routine configuration mirror in sync on every mutation (IMPL-R5).

Execution Side Effects

Scheduler ↔ CRUD sync (IMPL-R5). Each mutating endpoint calls a helper that mirrors the change into APScheduler:

- `_sched_add_routine(id)` — registers the job if the row is active and `execution_location != 'cloud'`.
- `_sched_update_routine(id)` — re-adds an active local job, otherwise removes it.
- `_sched_remove_routine(id)` — removes the job unconditionally.

All three swallow scheduler exceptions (logged, not propagated) so a jobstore fault cannot abort the underlying DB mutation.

Startup reconciliation (IMPL-R4). The `_init_scheduler_on_startup` startup hook:

1. Installs `_dispatch_routine_fire` via `set_dispatcher` .
2. Instantiates `ResmonScheduler` (with a disposable on-disk jobstore when the app DB is `:memory:` under tests).
3. Starts the scheduler.
4. Walks `get_routines(conn)` and calls `scheduler.add_routine(r)` for every row with `is_active` truthy; failures are logged per-row.

Routine fire pipeline (`_dispatch_routine_fire` , IMPL-R6). On every scheduled fire APScheduler calls `_routine_callback` , which delegates to the installed dispatcher. The dispatcher:

1. Loads the routine row; exits if missing or inactive.
2. Parses `parameters` JSON (unparseable payloads are logged and dropped).
3. Calls `admission.try_admit(kind="routine", routine_id=..., params_json=...)` ; returns early if the admission controller enqueues or drops the fire.
4. Loads `ai_settings` JSON, constructs a `SweepEngine` with `ai_enabled` and `ai_settings` , and calls `engine.prepare_execution("automated_sweep", repositories, params)` to reserve the execution row.
5. Stamps the `routine_id` column on the new `executions` row.
6. Registers the execution with `progress_store.register(exec_id)` and calls `_launch_execution(engine, exec_id, conn, ephemeral_credentials=None)` to run the sweep (the admission slot is released inside `_launch_execution` 's `finally` via `admission.note_finished`).
7. Updates `routines.last_executed_at = datetime('now')` .

APScheduler job options. Jobs are added with `replace_existing=True` , `coalesce=True` , and `misfire_grace_time=60` (see `ResmonScheduler.add_routine`), so overlapping missed fires collapse into one and late fires within 60 seconds still run.

Downstream effects of a fire. Depending on the routine's flags, a successful fire may additionally: write report artifacts under `resmon_reports/` , send an email via the SMTP pipeline (when `email_enabled` ; `email_ai_summary_enabled` inlines the AI summary), and raise a desktop notification (when `notify_on_complete` is ON and the global Settings → Notifications policy permits it).

Migration (IMPL-37). Local ⇄ cloud migration is orchestrated by the renderer in two steps; the two local endpoints (`released-to-cloud` , `adopt-from-cloud`) own only their half. Historical execution rows stay attached to the side that produced them.

Cloud path. Cloud routines are not registered with the local APScheduler; the cloud scheduler (`resmon-cloud` , per IMPL-32) owns their lifecycle. The desktop only reads and writes them through `cloudClient` .

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